Data and Algorithm Analysis

Chapter 21 — Minimum Spanning Trees

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Outline

MSTs

Generic Greedy Algorithms

Kruskal's Algorithm

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Definitions — MST

Let $G = (V, E)$ be an undirected, connected graph. A **spanning tree** of G is a set $T \subseteq E$ such that the subgraph (V, T) is acyclic and connected.

Question: If $|V| = n$, then what is $|T|$?

Let $w : E \to \mathbb{R}$ be a weight function on the edges of G. Then, the weight of T is

$$w(T) = \sum_{(u,v) \in T} w(u, v).$$

A **minimum spanning tree (MST)** of G is a spanning tree T of minimum weight.

Problem — MST

MINIMUM SPANNING TREE (MST)

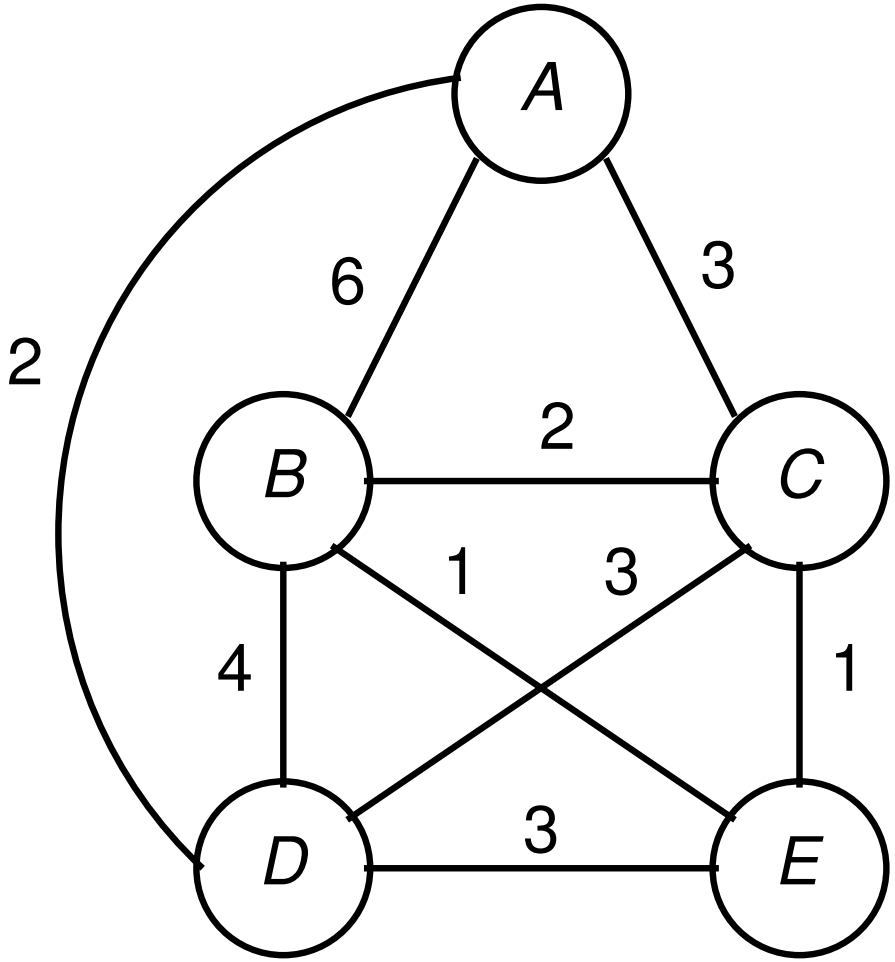
Input: Undirected, connected graph $G = (V, E)$; weight function $w : E \to \mathbb{R}$.

Output: Spanning tree $T \subseteq E$ of G such that

$$w(T) = \sum_{(u,v) \in T} w(u, v)$$

is minimum.

Example for MST



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Definitions — Safe Edges

Let $A \subseteq E$ be a set of edges that is contained in some MST of $G = (V, E)$. An edge $(u, v) \in E \setminus A$ is a **safe edge** for A if $A \cup \{(u, v)\}$ is contained in some MST.

Pseudocode — Generic Greedy MST

```
GENERIC-MST(G, w)
    A = ∅
    while A does not form a spanning tree
        select (u, v) ∈ E \ A that is safe for A
        A = A ∪ {(u, v)}
    return A
```

Definitions — Light Edges

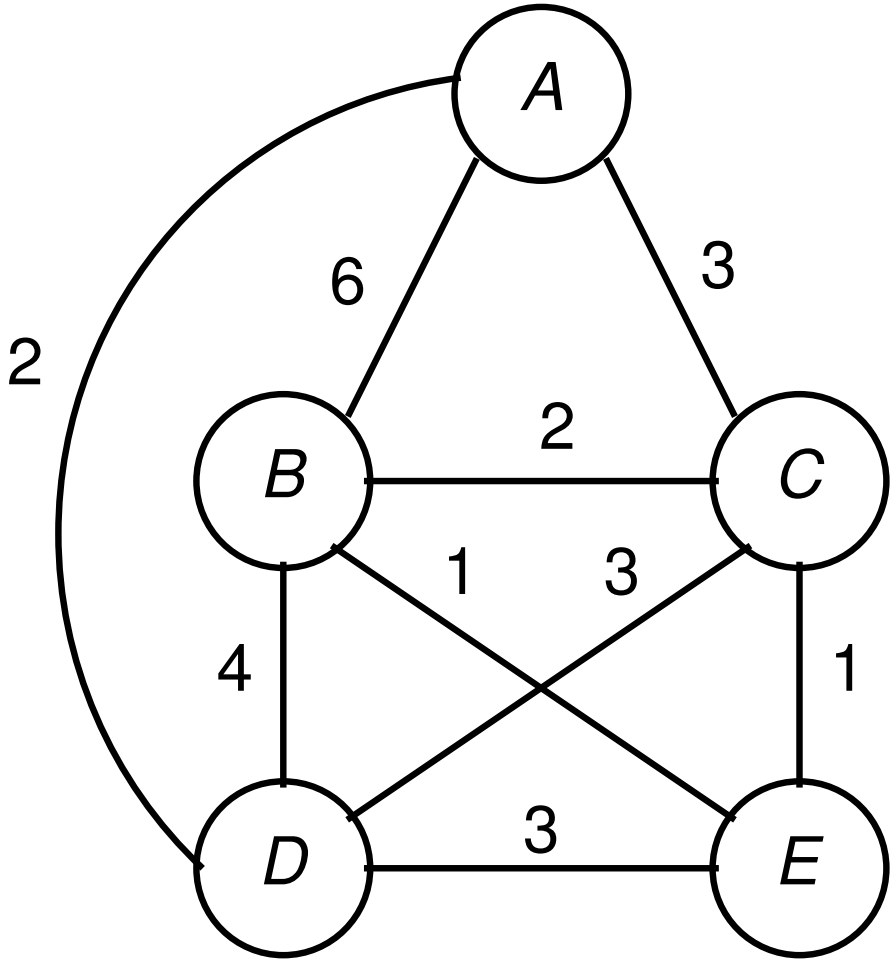
If $S \subseteq V$, $S \neq \emptyset$, and $S \neq V$, then the partition $(S, V \setminus S)$ of V is a **cut** of G.

An edge (u, v) **crosses the cut** $(S, V \setminus S)$ if $u \in S$ and $v \in V \setminus S$ or vice versa.

The cut $(S, V \setminus S)$ **respects** $A \subseteq E$ if no edge in A crosses the cut.

A **light edge** for a cut $(S, V \setminus S)$ is an edge crossing the cut of minimum weight.

Example for Cuts and Light Edges



Theorem 21.1

Theorem

Let $G = (V, E)$ be an undirected, connected graph with edge weight function $w : E \to \mathbb{R}$. Let $A \subseteq E$ be contained in some MST of G. Let $(S, V \setminus S)$ be any cut respecting A. Let (u, v) be a light edge crossing $(S, V \setminus S)$. Then, (u, v) is safe for A.

Proof of Theorem 21.1

Proof.
Let $T \subseteq E$ be an MST containing A. If $(u, v) \in T$, then (u, v) is safe for A, as desired. Otherwise, $(u, v) \notin T$. Then, construct another spanning tree T' of G as follows.
(See continuation of the proof on the board.) $\square$

Corollary 21.2

Corollary

Let $G = (V, E)$ be an undirected, connected graph with edge weight function $w : E \to \mathbb{R}$. Let $A \subseteq E$ be contained in some MST of G but not a spanning tree. Then, $F = (V, A)$ is a forest. Let $C = (V_C, E_C)$ be a connected component of F. If (u, v) is such that $u \in V_C$, $v \in V \setminus V_C$, and $w(u, v)$ is minimum, then (u, v) is safe for A.

Proof.

Just take $S = V_C$ in Theorem 21.1. □

Pseudocode — Light Edge MST

```
LIGHT-EDGE-MST(G, w)
    preprocessing to initialize data structure
    A = ∅
    while A does not form a spanning tree
        select a cut (S, V \ S) of G that respects A
        select (u, v) a light edge crossing (S, V \ S)
        A = A ∪ {(u, v)}
        update data structure
    return A
```

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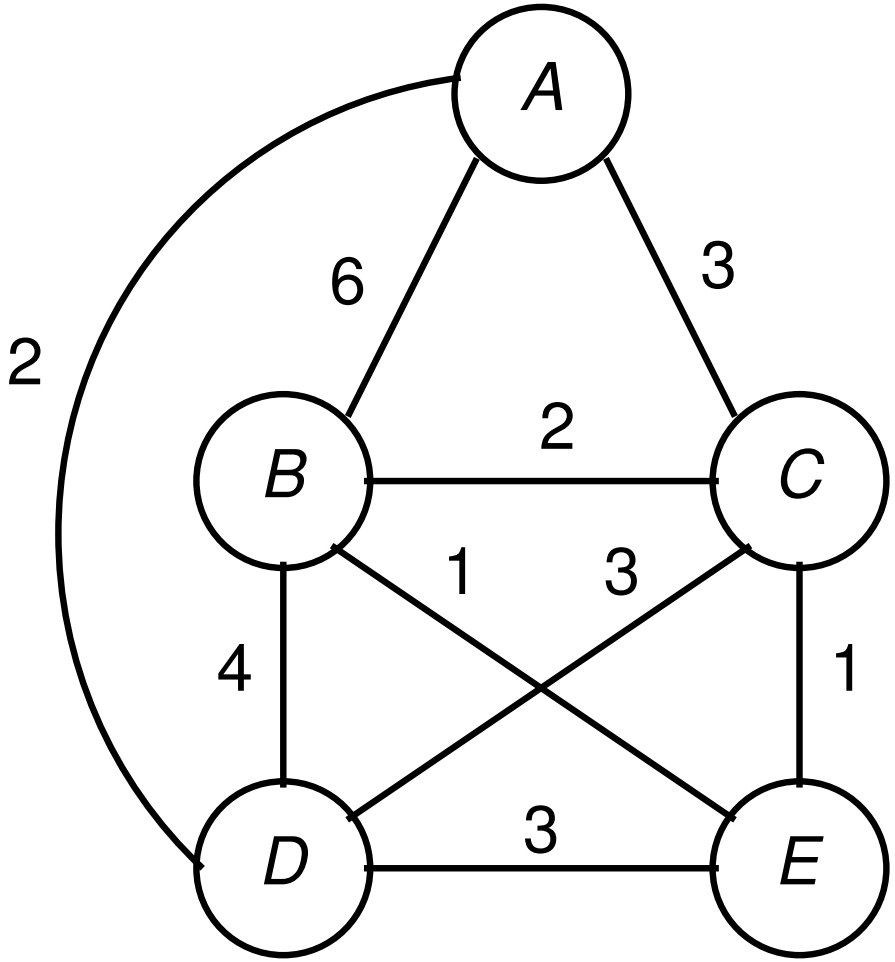
Intuition — Kruskal's Algorithm

- Process edges in nondecreasing order by weight and select an edge if it does not create a cycle.
- Maintain the connected components of the spanning forest to identify whether or not a cycle will be created.
- Use a data structure to efficiently maintain the connected components as a partition of V under these operations:
  - MAKE-SET(v)
  - UNION(u, v)
  - FIND(u)

Pseudocode — Kruskal's Algorithm

```
MST-KRUSKAL(G, w)
    A = ∅
    for v ∈ V
        MAKE-SET(v)
    sort E in nondecreasing order by weight w
    for (u, v) ∈ E in nondecreasing order by weight
        if FIND-SET(u) ≠ FIND-SET(v)
            A = A ∪ {(u, v)}
            UNION(u, v)
    return A
```

Example for Kruskal's Algorithm



Disjoint Sets Data Structure

Implementation: Disjoint set forest to represent a partition $\{V_1, V_2, \ldots, V_k\}$ of V. The vertices of each V_i are the vertices of a rooted tree with edges pointing toward the root and the **canonical element** at the root; each vertex v has a parent pointer $v.parent$.

- MAKE-SET(v): initially make each V_i a singleton tree, one for each $v \in V$.
- UNION(u, v): join two disjoint trees into a new tree.
- FIND(u): follow parent pointers to find the canonical element for the set.

Example on the board.

Time complexity of operations: ?

Time Complexity

Time complexity of Kruskal's Algorithm: ?

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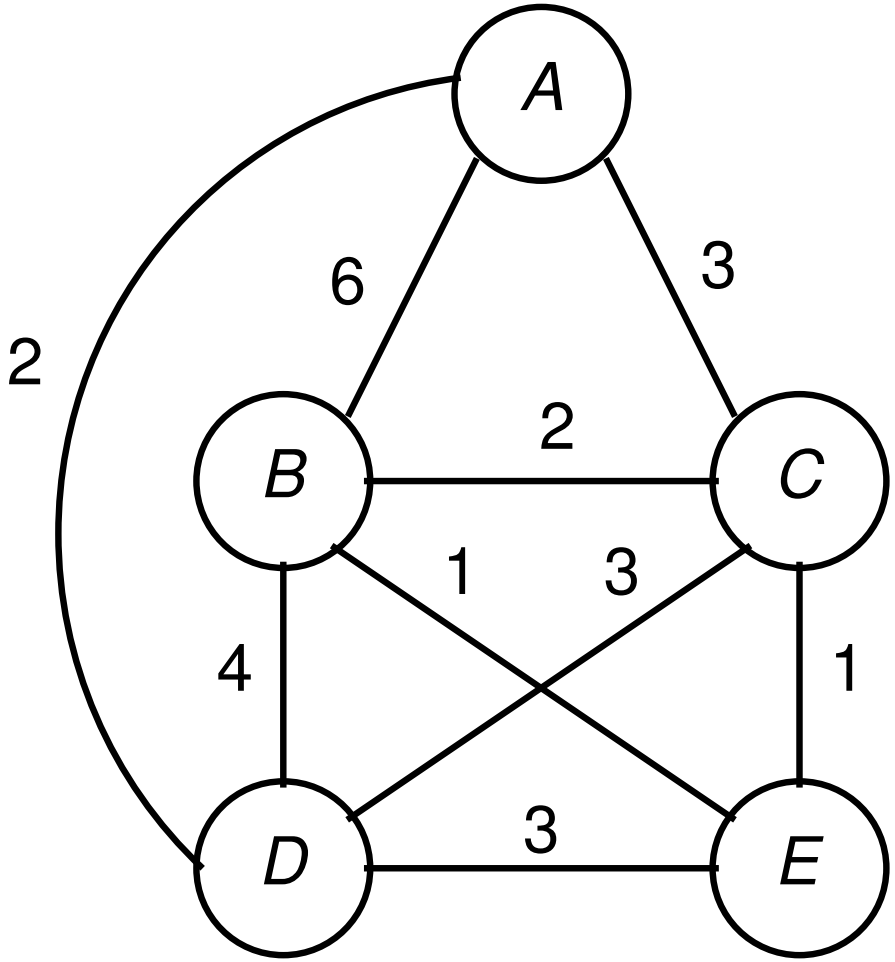
Intuition — Prim's Algorithm

- Grow a single tree in the spanning forest one edge at a time.
- The cut is from that the vertices spanned by that one tree to the rest of V.
- Use a **min priority** queue to choose a light edge crossing the cut.

Pseudocode — Prim's Algorithm

```
MST-PRIM(G, w, r)
for u ∈ V
    u.key = ∞
    u.π = NIL
r.key = 0
Q = V
while Q ≠ ∅
        u = EXTRACT-MIN(Q)
        for v adjacent to u
            if v ∈ Q and w(u, v) < v.key
                v.π = u
                v.key = w(u, v)
return u.π for u ∈ V
```

Example for Prim's Algorithm



Time Complexity

Time complexity of Prim's Algorithm: ?